

Vectors and Vector Spaces

Part 1/3: Foundations and Vector Operations

CSC 1105 Mathematics for Computer Science

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Lecture Outline

- Vectors, scalars and coordinate notation
- Vector addition, subtraction and scalar multiplication
- Vector length, distance and unit vectors
- Vector spaces and subspaces
- Applications in computer science

Module 4: Learning Across Three Parts

Part	Focus
1/3	Foundations: vectors, vector operations, vector spaces and subspaces.
2/3	Linear combinations, spanning sets, and an introduction to bases and dimension.
3/3	Linear dependence and independence, orthogonality, and related computations.

Learning Outcomes

By the end of this lecture, you should be able to:

- Represent vectors in $\mathbb{R}^2$, $\mathbb{R}^3$ and $\mathbb{R}^n$.
- Compute vector sums, differences, scalar multiples, lengths and distances.
- Explain the defining properties of a real vector space.
- Use the subspace test to classify simple subsets of $\mathbb{R}^n$.
- Interpret vector calculations in simple computing examples.

Connection to Matrix Algebra

Previously, we used matrices to transform vectors:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad A\mathbf{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

- The vector represents an input or output.
- The matrix represents a linear transformation.
- We now study vectors and the sets in which their operations take place.

Scalars and Vectors

Scalar

A scalar is a number used to scale a vector. In this part, scalars belong to $\mathbb{R}$.

Coordinate vector

A vector in $\mathbb{R}^n$ is an ordered list of n real numbers, usually written as a column.

$$a = -2, \quad \mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ -5 \end{bmatrix}.$$

The entries are the vector's **components**. Their order matters.

Vectors in Computer Science

Representation	Example	Meaning of components
Position	$(x, y, z)^T$	Coordinates of a 3D point
Colour	$(R, G, B)^T$	Red, green and blue intensities
Feature vector	$(x_1, \dots, x_n)^T$	Ordered measurements of an object
Signal segment	$(s_1, \dots, s_n)^T$	Samples at specified times

Modelling rule: every component must have a consistent interpretation across the vectors being compared.

Coordinate Spaces

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} : x_i \in \mathbb{R} \text{ for } i = 1, \dots, n \right\}.$$

- $\mathbb{R}^2$: vectors with two components, such as $(2, -1)^T$.
- $\mathbb{R}^3$: vectors with three components, such as $(2, -1, 4)^T$.
- $\mathbb{R}^n$: the same operations extend to any positive integer n .

The superscript n specifies the number of coordinates.

Row Vectors, Column Vectors and Equality

We use column vectors unless stated otherwise:

$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}, \quad \mathbf{u}^T = [2 \quad -1 \quad 4].$$

Two coordinate vectors are equal exactly when they have the same number of components and corresponding components are equal.

$$\begin{bmatrix} x \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ y \end{bmatrix} \iff x = 2, y = 3.$$

Order matters

$(2, 3)^T \neq (3, 2)^T$. A column and its transpose also have different matrix shapes.

Geometric Meaning of a Vector

In $\mathbb{R}^2$, the vector $\mathbf{u} = (3, 2)^T$ represents a displacement:

3 units horizontally and 2 units vertically.

Starting at the origin, this displacement ends at $(3, 2)$.

Starting at $P = (1, 4)$, the same displacement ends at

$$Q = (1 + 3, 4 + 2) = (4, 6).$$

Position and displacement

A point identifies a location. A displacement vector describes a change in location. The same displacement can start at different points.

Vector Addition

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, add corresponding components:

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}.$$

Example

$$\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 + 4 \\ -1 + 5 \\ 3 + (-2) \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix}.$$

Addition requires vectors from the same coordinate space.

Addition as Successive Displacements

Suppose a robot makes two displacements:

$$\mathbf{u} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Starting at $(0, 0)$, it visits $(3, 1)$ and then reaches

$$(3 + (-1), 1 + 2) = (2, 3).$$

Its total displacement is

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Geometrically, place the tail of the second displacement at the head of the first.

Zero Vector and Additive Inverse

In $\mathbb{R}^n$, the zero vector has n zero components:

$$\mathbf{0} = (0, \dots, 0)^T, \quad \mathbf{u} + \mathbf{0} = \mathbf{u}.$$

The additive inverse of $\mathbf{u}$ reverses every component:

$$-\mathbf{u} = (-u_1, \dots, -u_n)^T, \quad \mathbf{u} + (-\mathbf{u}) = \mathbf{0}.$$

For example, $-(2, -3)^T = (-2, 3)^T$.

Geometric meaning

The zero vector gives no displacement. The inverse gives a displacement of equal length in the opposite direction.

Subtraction and Displacement Between Points

Vector subtraction means addition of the inverse:

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v}).$$

Let $P = (1, 2)$ and $Q = (4, 6)$. The displacement from P to Q is

$$\overrightarrow{PQ} = \mathbf{q} - \mathbf{p} = \begin{bmatrix} 4 \\ 6 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

In the reverse direction,

$$\overrightarrow{QP} = \begin{bmatrix} -3 \\ -4 \end{bmatrix} = -\overrightarrow{PQ}.$$

Rule: destination coordinates minus starting coordinates.

Scalar Multiplication

For $a \in \mathbb{R}$ and $\mathbf{u} \in \mathbb{R}^n$,

$$a\mathbf{u} = \begin{bmatrix} au_1 \\ au_2 \\ \vdots \\ au_n \end{bmatrix}.$$

Example

$$-2 \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} -2(3) \\ -2(-1) \\ -2(4) \end{bmatrix} = \begin{bmatrix} -6 \\ 2 \\ -8 \end{bmatrix}.$$

Every component is multiplied by the same scalar.

Geometric Effect of Scaling

For a nonzero vector $\mathbf{u}$:

- $a > 0$: $a\mathbf{u}$ points in the same direction as $\mathbf{u}$.
- $a < 0$: $a\mathbf{u}$ points in the opposite direction.
- Its length is multiplied by $|a|$.
- $a = 0$: $a\mathbf{u} = \mathbf{0}$, which has no direction.

$$\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad 2\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \frac{1}{2}\mathbf{u} = \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}, \quad -\mathbf{u} = \begin{bmatrix} -2 \\ -1 \end{bmatrix}.$$

Example: Combined Vector Operations

Let $\mathbf{u} = (1, -2, 3)^T$ and $\mathbf{v} = (4, 0, -1)^T$. Find $2\mathbf{u} - 3\mathbf{v}$.

$$2\mathbf{u} = \begin{bmatrix} 2(1) \\ 2(-2) \\ 2(3) \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}, \quad 3\mathbf{v} = \begin{bmatrix} 3(4) \\ 3(0) \\ 3(-1) \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ -3 \end{bmatrix}.$$

Therefore,

$$2\mathbf{u} - 3\mathbf{v} = \begin{bmatrix} 2 - 12 \\ -4 - 0 \\ 6 - (-3) \end{bmatrix} = \boxed{\begin{bmatrix} -10 \\ -4 \\ 9 \end{bmatrix}}.$$

Check the signs: subtracting -3 gives $+3$.

Algebraic Rules for Vector Operations

For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$:

Rule	Identity
Commutativity	$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
Associativity of addition	$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
Distribution over vector addition	$a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
Distribution over scalar addition	$(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$
Compatibility of scaling	$a(b\mathbf{u}) = (ab)\mathbf{u}$
Scalar identity	$1\mathbf{u} = \mathbf{u}$

Exercise: Vector Arithmetic

Let $\mathbf{u} = (2, -1)^T$, $\mathbf{v} = (-3, 4)^T$ and $\mathbf{w} = (1, 2)^T$.

- 1 Compute $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$.
- 2 Compute $3\mathbf{u} + 2\mathbf{v}$.
- 3 Find $\mathbf{x}$ such that $\mathbf{x} + \mathbf{u} = \mathbf{w}$.
- 4 Is $(2, -1)^T + (1, 0, 3)^T$ defined in ordinary vector addition? Explain.

Vector Length: The Euclidean Norm

For $\mathbf{u} \in \mathbb{R}^n$, its Euclidean norm is

$$\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}.$$

In $\mathbb{R}^2$, this is the Pythagorean formula. For example,

$$\left\| \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$

- $\|\mathbf{u}\| \geq 0$, with equality exactly when $\mathbf{u} = \mathbf{0}$.
- $\|a\mathbf{u}\| = |a| \|\mathbf{u}\|$.

Distance Between Two Vectors

The Euclidean distance between $\mathbf{u}$ and $\mathbf{v}$ is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{\sum_{i=1}^n (u_i - v_i)^2}.$$

For $\mathbf{u} = (1, 2, 3)^T$ and $\mathbf{v} = (4, 6, 3)^T$,

$$\mathbf{u} - \mathbf{v} = \begin{bmatrix} -3 \\ -4 \\ 0 \end{bmatrix}, \quad d(\mathbf{u}, \mathbf{v}) = \sqrt{9 + 16 + 0} = 5.$$

Interpretation

Distance measures the length of the difference vector. It is zero exactly when the two vectors are equal.

Unit Vectors and Normalization

A **unit vector** has length 1. For $\mathbf{u} \neq \mathbf{0}$, define

$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}.$$

For $\mathbf{u} = (3, 4)^T$,

$$\hat{\mathbf{u}} = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}, \quad \|\hat{\mathbf{u}}\| = \sqrt{\frac{9}{25} + \frac{16}{25}} = 1.$$

Normalization preserves direction while making the length equal to 1.

Zero vector

The zero vector cannot be normalized because division by its norm would mean division by zero.

Application: Comparing Feature Vectors

Suppose a document has two features: counts of two selected words.

$$\mathbf{q} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}.$$

Distances from the query vector are

$$d(\mathbf{q}, \mathbf{a}) = \sqrt{(2 - 3)^2 + (1 - 1)^2} = 1,$$

$$d(\mathbf{q}, \mathbf{b}) = \sqrt{(2 - 0)^2 + (1 - 4)^2} = \sqrt{13}.$$

Under this representation and distance, $\mathbf{a}$ is closer to $\mathbf{q}$.

Feature choices and scales affect the meaning of closeness.

Beyond Individual Vectors

Consider the set of all vectors on the horizontal axis:

$$W = \left\{ \begin{bmatrix} x \\ 0 \end{bmatrix} : x \in \mathbb{R} \right\} \subseteq \mathbb{R}^2.$$

Adding two members keeps us in the set:

$$\begin{bmatrix} x \\ 0 \end{bmatrix} + \begin{bmatrix} y \\ 0 \end{bmatrix} = \begin{bmatrix} x + y \\ 0 \end{bmatrix} \in W.$$

Scaling any member also keeps us in the set:

$$a \begin{bmatrix} x \\ 0 \end{bmatrix} = \begin{bmatrix} ax \\ 0 \end{bmatrix} \in W.$$

These are the two **closure properties** underlying vector spaces.

Real Vector Spaces

Definition

A real vector space is a nonempty set V with vector addition and multiplication by real scalars, satisfying the vector-space axioms.

- Addition takes $\mathbf{u}, \mathbf{v} \in V$ to $\mathbf{u} + \mathbf{v} \in V$.
- Scalar multiplication takes $a \in \mathbb{R}$, $\mathbf{u} \in V$ to $a\mathbf{u} \in V$.
- The axioms specify how these operations must behave.

Vectors may be coordinate lists, matrices, polynomials or other objects.

Vector-Space Axioms: Addition

For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$:

- 1 **Closure:** $\mathbf{u} + \mathbf{v} \in V$.
- 2 **Commutativity:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
- 3 **Associativity:** $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
- 4 **Zero vector:** there exists $\mathbf{0} \in V$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
- 5 **Additive inverse:** for each $\mathbf{u}$, there exists $-\mathbf{u} \in V$ with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.

Vector-Space Axioms: Scalar Multiplication

For all $\mathbf{u}, \mathbf{v} \in V$ and $a, b \in \mathbb{R}$:

- ⑥ **Closure:** $a\mathbf{u} \in V$.
- ⑦ **Distribution over vectors:** $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$.
- ⑧ **Distribution over scalars:** $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$.
- ⑨ **Compatibility:** $a(b\mathbf{u}) = (ab)\mathbf{u}$.
- ⑩ **Identity:** $1\mathbf{u} = \mathbf{u}$.

Together with the addition axioms, these define a real vector space.

Why $\mathbb{R}^n$ Is a Vector Space

Componentwise addition and scaling produce real components, so both operations are closed in $\mathbb{R}^n$.

The remaining properties follow from real-number arithmetic. For example,

$$[a(\mathbf{u} + \mathbf{v})]_i = a(u_i + v_i) = au_i + av_i = [a\mathbf{u} + a\mathbf{v}]_i.$$

This holds for every component i , so

$$a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}.$$

The zero vector and additive inverses are also defined componentwise.

Other Examples of Vector Spaces

Matrices of a fixed size

$M_{m \times n}(\mathbb{R})$ is the set of all real $m \times n$ matrices, with ordinary matrix addition and scalar multiplication. Its zero vector is the zero matrix.

Polynomials of degree at most two

$P_2 = \{a + bx + cx^2 : a, b, c \in \mathbb{R}\}$, with ordinary polynomial addition and scalar multiplication. This set includes the zero polynomial.

For example,

$$(1 + 2x) + (3 - x + x^2) = 4 + x + x^2 \in P_2.$$

The defining operations and axioms determine whether a set is a vector space.

A Set of Vectors May Fail to Be a Vector Space

Consider valid RGB colour vectors:

$$C = \left\{ \begin{bmatrix} r \\ g \\ b \end{bmatrix} : r, g, b \in \{0, 1, \dots, 255\} \right\}.$$

With ordinary operations,

$$\begin{bmatrix} 200 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 300 \\ 0 \\ 0 \end{bmatrix} \notin C, \quad -1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \notin C.$$

Therefore C is not a real vector space under these operations.

Representation and algebra

RGB values lie in $\mathbb{R}^3$, but the restricted set of valid colours is not closed under arbitrary addition and real scaling.

Subspaces

Definition

A subset W of a vector space V is a subspace if W is itself a vector space using the same addition and scalar multiplication as V .

- The ambient space V supplies the operations.
- A subspace must contain the zero vector of V .
- Both $\{\mathbf{0}\}$ and V itself are subspaces of V .

A line through the origin in $\mathbb{R}^2$ is a typical example.

The Subspace Test

Let $W \subseteq V$, where V is already a vector space. Check:

- 1 **Zero vector:** $\mathbf{0} \in W$.
- 2 **Closure under addition:** for arbitrary $\mathbf{u}, \mathbf{v} \in W$, show that $\mathbf{u} + \mathbf{v} \in W$.
- 3 **Closure under scalar multiplication:** for arbitrary $a \in \mathbb{R}$ and $\mathbf{u} \in W$, show that $a\mathbf{u} \in W$.

Why these checks suffice

The other algebraic identities hold in V and carry over to W . Closure under scaling supplies additive inverses by taking $a = -1$.

Example: A Line Through the Origin

Let $W = \{(x, y)^T \in \mathbb{R}^2 : y = 2x\}$. Every member has the form $(t, 2t)^T$.

① **Zero:** $t = 0$ gives $(0, 0)^T \in W$.

② **Addition:** for any $s, t \in \mathbb{R}$,

$$\begin{bmatrix} s \\ 2s \end{bmatrix} + \begin{bmatrix} t \\ 2t \end{bmatrix} = \begin{bmatrix} s+t \\ 2(s+t) \end{bmatrix} \in W.$$

③ **Scaling:** for any $a, t \in \mathbb{R}$,

$$a \begin{bmatrix} t \\ 2t \end{bmatrix} = \begin{bmatrix} at \\ 2(at) \end{bmatrix} \in W.$$

Therefore W is a subspace of $\mathbb{R}^2$.

Example: A Homogeneous Constraint

Let $W = \{(x, y, z)^T \in \mathbb{R}^3 : x + y + z = 0\}$.

- The zero vector belongs to W because $0 + 0 + 0 = 0$.
- If $\mathbf{u}, \mathbf{v} \in W$, then

$$(u_1 + v_1) + (u_2 + v_2) + (u_3 + v_3) = (u_1 + u_2 + u_3) + (v_1 + v_2 + v_3) = 0.$$

Thus $\mathbf{u} + \mathbf{v} \in W$.

- For any scalar a ,

$$au_1 + au_2 + au_3 = a(u_1 + u_2 + u_3) = 0.$$

Thus $a\mathbf{u} \in W$.

Therefore W is a subspace of $\mathbb{R}^3$.

Connection to Kernels

For an $m \times n$ matrix A , recall

$$\ker(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}.$$

- $A\mathbf{0} = \mathbf{0}$, so the zero vector belongs to the kernel.
- If $A\mathbf{u} = \mathbf{0}$ and $A\mathbf{v} = \mathbf{0}$, then $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0}$.
- If $A\mathbf{u} = \mathbf{0}$, then $A(a\mathbf{u}) = aA\mathbf{u} = \mathbf{0}$.

Therefore $\ker(A)$ is always a subspace of $\mathbb{R}^n$.

Exercise: Subspaces

Under the usual operations, which sets are subspaces of $\mathbb{R}^2$?

- ① $W_1 = \{(x, y)^T : y = -x\}$.
- ② $W_2 = \{(x, y)^T : x \geq 0, y \geq 0\}$.
- ③ $W_3 = \{(x, y)^T : xy = 0\}$, the union of the coordinate axes.
- ④ $W_4 = \{(x, y)^T : x + y = 1\}$.

For a positive answer, verify the subspace test. For a negative answer, give a counterexample or a failed condition.

Solutions: Subspace

- ① W_1 is a subspace: it contains $\mathbf{0}$, and

$$(s, -s)^T + (t, -t)^T = (s + t, -(s + t))^T, \quad a(t, -t)^T = (at, -at)^T.$$

- ② W_2 fails scalar closure: $(1, 1)^T \in W_2$, but $-(1, 1)^T = (-1, -1)^T \notin W_2$.
- ③ W_3 fails addition: $(1, 0)^T, (0, 1)^T \in W_3$, but their sum $(1, 1)^T \notin W_3$.
- ④ W_4 does not contain $\mathbf{0}$, since $0 + 0 \neq 1$.

Common Mistakes: Warning!

- Adding vectors with different numbers of components.
- Treating $\|\mathbf{u}\|$ as a vector. A norm is a nonnegative scalar.
- Attempting to normalize the zero vector.
- Concluding that a set is a subspace only because it contains $\mathbf{0}$.
- Checking a few examples to prove closure. A proof must cover arbitrary members and scalars.

Revision Questions

- 1 For $\mathbf{u} = (1, 2, -2)^T$ and $\mathbf{v} = (3, 0, 1)^T$, find $2\mathbf{u} - \mathbf{v}$, $\|\mathbf{u}\|$, $d(\mathbf{u}, \mathbf{v})$ and $\hat{\mathbf{u}}$.
- 2 Prove that $W = \{(x, y, z)^T : x - 2y + z = 0\}$ is a subspace of $\mathbb{R}^3$.
- 3 Explain why the polynomials of degree exactly two do not form a vector space under the usual operations.
- 4 Find real numbers a, b such that

$$a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Questions?